

Dexterity from Supervised Play

Anonymous Authors

Abstract—How can robots autonomously collect the contact-rich experience needed for dexterous manipulation? We present **DSP: Dexterity from Supervised Play**, a framework in which the robot learns a goal-agnostic play policy from minutes of human play, deploys it to collect hours of interaction data to train an action-conditioned world model. At inference time, the play policy also proposes candidate action sequences and model-predictive control (MPC) selects among their predicted outcomes for a requested goal. In simulated any-to-any cube reorientation, our DSP play-and-model pipeline reaches within 0.6 rad on 91.4% of trials and within 0.4 rad on 72.7% using only $\sim 700k$ frames, compared with 53.9% and 25.8% for hindsight behavior cloning; RL with on-policy reaches 86.7% at both thresholds only after 344M counted frames. On real-robot data, increasing the nested autonomous-play corpus from 0.54 to 2.34 hours reduces held-out position and rotation prediction error by 12.1% and 14.0%, respectively. These results show that supervised play can scale autonomous data collection and provide a reusable action prior for model-based dexterous control without task-specific demonstrations.

I. INTRODUCTION

Humans, especially infants and children, develop dexterity from playing [1]. Play offers a task-agnostic source of interaction. Similarly, for robots, learning via playing holds tremendous potential for learning a rich physical understanding of the world required for dexterous manipulation without human supervision. In practice, however, play is a difficult behavior to engineer.

Prior works model *play* by manually defining explicit heuristics as guidance [2], [3], [4] and by engineering restrictive safety boundaries to prevent hardware damage [5]. However, play is inherently intuitive and difficult to formalize with hand-crafted rules, while manual safety constraints require excessive tuning that limits scalability.

In this work, our core hypothesis is that play is a behavior that can be modeled as a ‘goal-agnostic’ state-conditioned probabilistic model in a Markov Decision Process (MDP) and therefore be learned through a standard behavior cloning pipeline. In other words, even though a safe play behavior is hard to be defined explicitly with heuristics, it can easily be learned implicitly through demonstration data.

Therefore, we propose DSP: *Dexterity from Supervised Play*, a framework that turns minutes of human play into a policy that can continue collecting robot interaction autonomously. An action-conditioned world model learns from these data, while the same play policy provides candidate action sequences for model-predictive control (MPC). As a preliminary acceleration study, we also explore distilling a closed-loop, goal-conditioned policy through self-play within the learned world model.

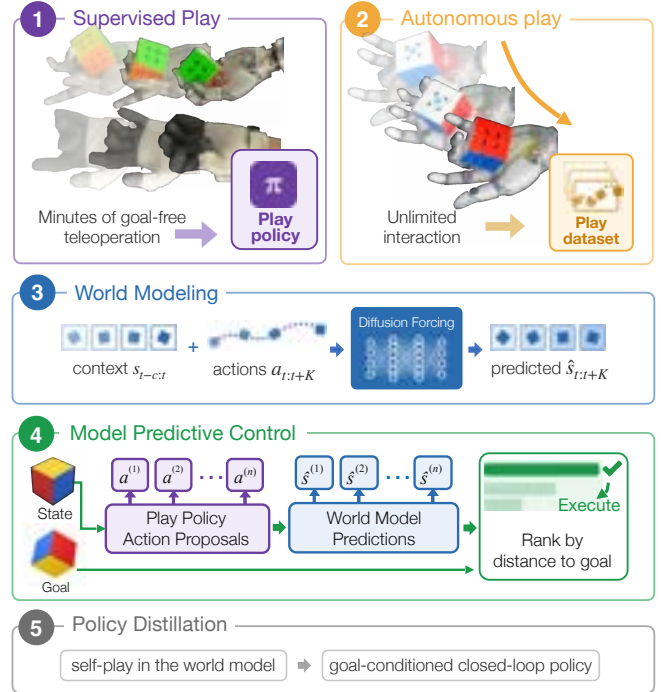


Fig. 1. **Method overview.** [Akash: Task goal enters here, no need of ‘only’] Limited human play trains a supervised play policy. Autonomous policy rollouts produce contact-rich data for an action-conditioned world model. A downstream goal is introduced only when MPC scores play-policy proposals. Distillation is preliminary and is not required for the core result.

Our primary contribution is a framework for learning dexterous manipulation from data collected by an autonomous play policy. We evaluate how autonomous play supports world-model learning and goal-directed planning through four technical contributions:

- **Supervised play.** We formulate supervised play as a mechanism for turning a limited human interaction prior into autonomous contact-rich data collection.
- **Scaling autonomous play data.** We show that held-out world-model prediction improves monotonically as the real-robot autonomous-play dataset grows from 0.54 to 2.34 hours.
- **Play policy as an action prior.** We show that model-based selection extracts substantially more goal-directed value from play-policy proposals than direct play or hindsight BC.
- **Model-stack improvement in control.** Holding the play policy, planner, goals, and evaluation seeds fixed, we show that a newer world-model stack improves arbitrary-goal acquisition.

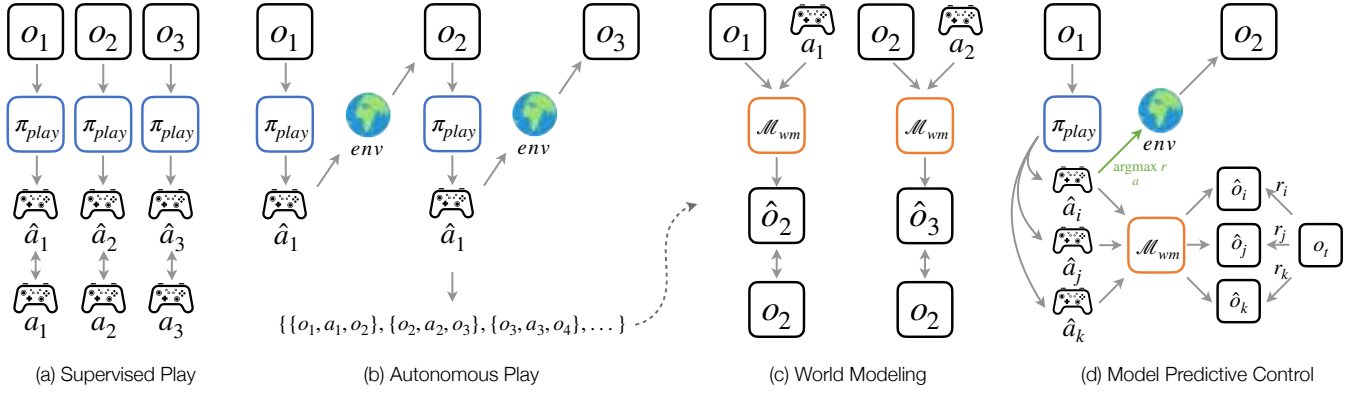


Fig. 2. **Dexterity from Supervised Play.** (a) A goal-agnostic play policy is trained by behavior cloning on human teleoperation data. (b) The learned policy interacts autonomously with the environment to collect observation–action transitions. (c) These transitions train an action-conditioned world model to predict future observations. (d) At inference time, the play policy proposes candidate actions, the world model predicts their outcomes, and model-predictive control selects the candidate with the highest goal-conditioned reward. The selected action is executed in the environment, and planning repeats from the new observation.

In simulated arbitrary-goal reorientation, our 701k-frame controller reaches within 0.6 rad on 91.4% of trials and within 0.4 rad on 72.7%, compared with 53.9% and 25.8% for hindsight BC. PPO-Match reaches 86.7% at both thresholds after 344M counted frames, about 490 times the counted-frame budget, so it trails our controller at 0.6 rad and leads it at 0.4 rad; PPO-Best is retained as the stronger task-specific precision and drop-avoidance reference. Increasing the nested real-robot play corpus from 0.54 to 2.34 hours reduces held-out position and rotation prediction error by 12.1% and 14.0%. We further specify matched analyses of play-policy architecture and world-model design, and report a provisional hardware threshold-reach summary whose protocol is being frozen. The complete evaluation also includes Push-T and required RGB reorientation experiments; their reserved tables will be filled from the frozen runs. The code, data, and training details will be released for reproducibility.

II. RELATED WORK

A. Play and interaction in RL

RL agents can acquire sophisticated behavior via open-ended interaction and planning. Model-based agents such as MuZero [6] learn a latent dynamics model and plan within it to master board games and Atari without access to the rules. In multi-agent settings, emergent autocurricula produce complex strategies purely from self-play and environment interaction, as in the hide-and-seek tool-use environment [7]. Comparable interaction-driven pipelines now reach elite performance on physical systems such as competitive table tennis [5]. These results establish that interaction and planning are powerful, but they depend on well-specified rewards, simulators, or game rules. DSP instead targets contact-rich manipulation, where reward engineering and designing safe exploration plans are difficult, and obtains its interaction behavior from human play rather than from reward-driven search.

B. Curiosity and intrinsic motivation

To reduce dependence on hand-designed rewards, intrinsic-motivation methods reward the agent for reaching novel or hard-to-predict states. Prediction-error curiosity uses the error of a learned forward as an intrinsic signal [3], [8], while Random Network Distillation measures novelty as the error of predicting a fixed random network’s features [9]. While such signals drive broad exploration, they are still heuristics and they do not bias interactions toward safe, object-centric contact that dexterous manipulation requires.

C. Learning from play

Closest to our framing is robot learning from play: goal-free, human-generated interaction used as a task-agnostic source of supervision. Play-LMP learns latent plans from teleoperated play and reuses them for goal-conditioned control [10], and subsequent work distills affordances from play to improve sample efficiency [4]. These approaches largely treat play as a fixed dataset for imitation. In contrast, DSP uses a human play data seed to train a policy that serves as a reusable, autonomous data-collection mechanism.

D. Dexterous manipulation

Dexterous, multi-fingered manipulation has been approached both by model-free reinforcement learning and by imitation. RL methods learn agile in-hand skills such as general object reorientation [11], [12], [13] and sim-to-real cube manipulation [14], [15], but require carefully shaped task rewards and hundreds of millions of simulated interactions. DSP differs from existing work and does not use simulation data: it learns contact-rich dynamics purely from autonomous play, without task-specific demonstrations or rewards, and plans for arbitrary goals by scoring play-policy proposals with the learned world model under MPC.

III. METHOD

A. Overview

Given a dexterous manipulation task, our pipeline DSP achieves it in 4 stages described in Fig. 2. We first collect

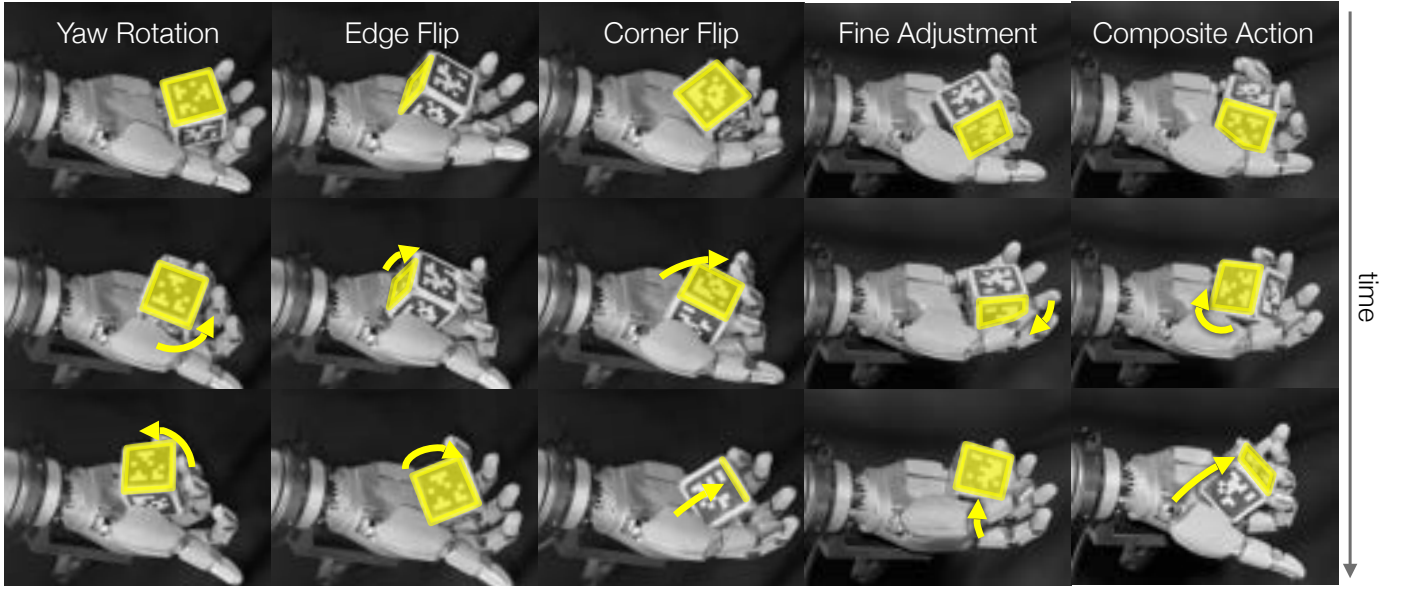


Fig. 3. Atomic skills. XXX

data through teleoperation to train a play policy (see Sec. III-B). We then rollout this policy on the robot autonomously to collect a play dataset that contains observation–action transitions (see Sec. III-C). This play dataset is used to train an action-conditioned world model to model the dynamics of the interaction (see Sec. III-D). For inference, we could either directly perform online MPC to predict optimal action sequences (see Sec. III-E) or optionally distill a goal-conditioned closed-loop policy from self-play to amortize the inference cost (see Sec. III-F).

B. Supervised Play

Play is difficult to engineer but intuitive for humans. We therefore train a diffusion-transformer [16] play policy by behavior cloning on goal-free human teleoperation $\mathcal{D}_{\text{human}}$ (Fig. 2(a)). Let o_t denote the observation of the robot and object and a_t a recorded joint command. The policy samples action chunks $\hat{a}_{t:t+H-1}$ conditioned on observation history:

$$\hat{a}_{t:t+H-1} \sim \pi_{\text{play}}(\cdot \mid o_{\leq t}), \quad (1)$$

where H is the action horizon. The demonstrations capture continued object interaction without prescribed goals or task-specific rewards. Safe behavior are not explicitly engineered but implicitly demonstrated in the data.

C. Autonomous Play

We then roll out the policy autonomously on the robot to collect a play dataset $\mathcal{D}_{\text{play}}$ which achieves sufficient coverage in the state-transition space as described in Fig. 2b. This autonomous data collection process requires little human supervision and can be parallelized across multiple robots and environments. We record sensory and proprioceptive observations as transitions (o_t, a_t, o_{t+1}) .

D. World Modeling

With the play dataset collected from Sec. III-C, we train a world model to predict future observations conditioned on recent observation history and an input action chunk:

$$\hat{o}_{t+1:t+K} = \mathcal{M}_{\text{wm}}(o_{t-C+1:t}, a_{t:t+K-1}). \quad (2)$$

Here, C and K denote the context length and prediction horizon, respectively. The training objective has the generic form

$$\mathcal{L}(\phi) = \mathbb{E}_{(o,a) \sim \mathcal{D}_{\text{play}}} \left[\sum_{k=1}^K \lambda_k \ell(\hat{o}_{t+k}, o_{t+k}) \right], \quad (3)$$

where ϕ denotes the parameters of $\mathcal{M}_{\text{wm}}$. We train our action-conditioned world model using Diffusion Forcing [17], which assigns independent noise levels to observations at different time steps and learns to denoise them causally. This allows the model to condition on clean observed history while progressively denoising future observations, supporting autoregressive rollouts under candidate action sequences.

E. Model Predictive Control

We combine the learned play policy π_{play} and world model $\mathcal{M}_{\text{wm}}$ for MPC (Fig. 2d). The play policy provides an action prior that focuses search on demonstrated interaction behaviors. At time t , we sample N candidate action chunks and predict their outcomes over $K \leq H$ steps:

$$\begin{aligned} \hat{a}_{t:t+H-1}^{(i)} &\sim \pi_{\text{play}}(\cdot \mid o_{\leq t}), \\ \hat{o}_{t+1:t+K}^{(i)} &= \mathcal{M}_{\text{wm}}(o_{t-C+1:t}, \hat{a}_{t:t+K-1}^{(i)}), \end{aligned} \quad (4)$$

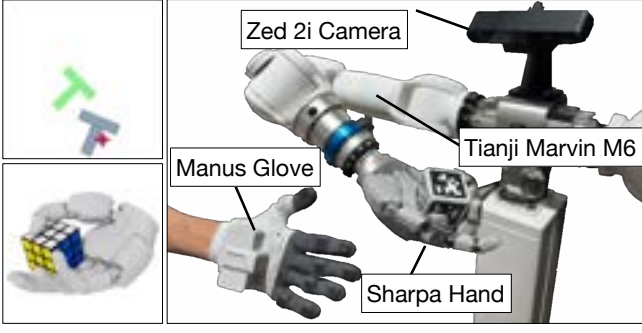


Fig. 4. **Experiment setup.** [TODO: Fill the control rate, object dimensions and mass, compute, calibration, safety limits, tracker filtering, and reset protocol.]

for $i = 1, \dots, N$. Given a target observation o_g , we score each candidate by its best predicted goal reward and select

$$\begin{aligned} r_i &= \max_{1 \leq k \leq K} r(\hat{o}_{t+k}^{(i)}, o_g), \\ i^* &= \arg \max_{1 \leq i \leq N} r_i. \end{aligned} \quad (5)$$

Here, $r(o, o_g)$ measures how well observation o satisfies the goal o_g . We execute the first E actions ($1 \leq E \leq K$) of the selected chunk, receive a new observation, and replan. The goal thus enters through the reward, while the play policy and world model remain goal-agnostic.

F. Policy Distillation

MPC incurs inference latency from repeated candidate evaluation. We therefore propose optional policy distillation through self-play in the learned world model. Starting from recorded observation histories and sampled goals o_g , we run MPC within $\mathcal{M}_{\text{wm}}$, advancing the history with model predictions after each selected action prefix. This produces a dataset $\mathcal{D}_{\text{distill}}$ of tuples $(h, o_g, \hat{a}^*)$, where h is the current observation history and $\hat{a}^* = \hat{a}_{t:t+E-1}^{(i^*)}$ is the prefix selected by MPC. We train a goal-conditioned policy π_θ by behavior cloning:

$$\mathcal{L}_{\text{distill}}(\theta) = -\mathbb{E}_{\mathcal{D}_{\text{distill}}} [\log \pi_\theta(\hat{a}^* | h, o_g)]. \quad (6)$$

At deployment, π_θ directly predicts an action prefix from the actual observation history and goal. It executes the prefix and updates its prediction from new observations, yielding a closed-loop policy that amortizes MPC search.

IV. EXPERIMENTS

In the experiments section, we aim to evaluate three questions. First, can goal-agnostic play and world modeling enable goal-conditioned manipulation without task-specific demonstrations, in both simulation and real world? Second, how does play-guided model-predictive control (MPC) compare with direct policy learning and task-specific reinforcement learning? Third, how do the design of play policy and the scale of play data affect prediction and control?

A. Experimental Setup

a) *Tasks.*: We organize the evaluation around four tasks. **Task A: Push-T** tests state-based planar pushing to randomized target poses and measures success and best mask overlap. **Task B: simulated reorientation** asks a multifingered hand to move a cube between initial and goal orientations separated by more than 90° . **Task C: real reorientation** evaluates the corresponding hardware problem with tag-tracked cube pose. **Task D: RGB reorientation** is the required visual-input experiment: it uses the same reorientation goals and evaluation protocol as Task C while removing tracked object pose from the deployed controller observation. Push-T and RGB result cells remain placeholders until their frozen evaluations are inserted.

For reorientation, the task error and goal constraint are

$$e_t = d_{\text{SO}(3)}(R_t, R_g), \quad e_0 > 90^\circ. \quad (7)$$

For Task B, we sample initial and goal orientations uniformly from $\text{SO}(3)$ subject to Eq. (7). This removes short-displacement trials included by unconstrained random-goal protocols.

b) *Simulation setup.*: We use Isaac Lab as the simulator. Each Task B controller is evaluated on the same 128 seeded trials for 2,500 control frames. Dropped cubes are respawned and the trial continues. The robot observes hand state and cube pose and commands 22 joint positions. The world-model planner samples 1,024 play-policy proposals, predicts 50 steps and executes 25 steps. In Task D, the controller instead receives the RGB stream; tag-derived pose is reserved for external evaluation and any use for training supervision must be reported explicitly.

c) *Real setup.*: As shown in Fig. 4, the hardware platform uses a fixed Sharpa right hand with 22 commanded joints, a Manus glove collecting teleoperation data for human play, a ZED 2i camera, and cube-mounted tags for six-degree-of-freedom pose tracking. Logged signals include proprioception and tracked cube pose. As the real world controller runs slower than the simulation, we increase the time budget to 120 seconds.

B. Baselines

a) *Simulation.*: We compare with: (1) the *play policy* executed without a task goal; (2) *hindsight BC*, trained from the same 701K transitions as our archived MPC system after relabeling achieved outcomes; (3) *goal-conditioned BC*, a policy trained from 27 teleoperated takes that each rotate the cube toward one reference pose and are truncated at their closest approach, scored here on the same arbitrary-goal protocol as every other row; (4) *PPO-TM* (timesteps matched), PPO trained on the same 701K-frame budget as Ours/hindsight BC; (5) *PPO-PM* (performance matched), the checkpoint at 344M frames with comparable SR@0.4 rad; and (6) *PPO-unlimited*, the median of five checkpoints from 360M frames onward as a stronger reference. Note that PPO-unlimited represents the theoretical RL limit, which is either impossible to learn directly in the real world or subject to the sim-to-real gap. Our method uses the play policy as a

TABLE I
PUSH-T RESULTS.

Controller	SR@90	SR@95	Coverage
Behavior cloning	64	28	0.751
Hindsight BC	74	36	0.764
Goal-conditioned BC	60	34	0.733
RL, matched data	0	0	0.155
Ours	98	74	0.953

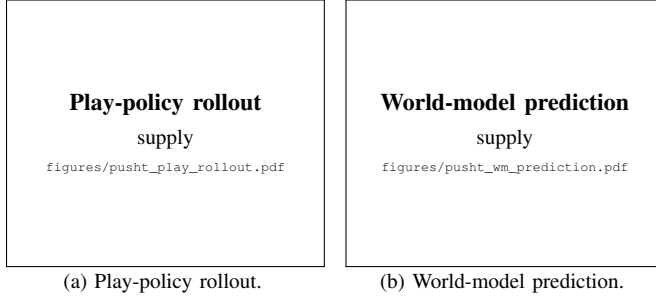


Fig. 5. **Push-T qualitative results.** (a) A rollout of the play policy. (b) World-model predictions for the sampled proposals from the same state; the trajectory ranked best by MPC is drawn in green.

proposal distribution and scores proposals with the learned world model.

b) Real.: The real-world experiment tests whether our method transfers to hardware while retaining near-simulation performance. We compare with joint random walk, cube random walk on $SO(3)$ with the same average angular velocity as the play policy, and the play policy. The remaining simulation baselines are not transferable to hardware. For example, RL requires millions of interaction frames, and its unstable early exploration can damage the hardware.

C. Results

1) Task A: Push-T: Push-T is intended to test whether one goal-agnostic play dataset and world model can be reused for held-out planar goals. Figure 5 shows a play-policy rollout next to the world-model predictions that MPC ranks. The final comparison will use BC, hindsight BC, goal-conditioned BC, interaction-matched RL, converged RL, and play-guided MPC under one protocol. [TODO: Fill the action space, goal distribution, data budget, episode horizon, success threshold, repeated seeds, final IoU, SE(2) error, and held-out-goal result.]

2) Task B: Simulated Distant-Goal Reorientation: Figure 6 reports the primary Task B metrics as SR@0.4/0.6/0.8 rad on 128 distant-goal trials ($e_0 > 90^\circ$). Every value is counted directly from the per-trial closest-approach of the frozen rollouts, not interpolated from the archived degree thresholds. All controllers share one protocol: start orientation and goal drawn uniformly from $SO(3)$ at least 90° apart, 2500 control frames, and a dropped cube respawned in the hand. Play-guided MPC converts the goal-agnostic play policy’s 15.6% SR@0.4 rad into 72.7%, and hindsight relabeling of the same 701K transitions recovers only about

a fifth of that gap (25.8%). Goal-conditioned BC, trained toward a single reference pose and given no goal input, does not transfer to arbitrary goals: it is marginally more precise than the play policy at the tightest threshold (21.9% versus 15.6%) and worse at the two coarser ones, which is the signature of a policy that rotates the cube without steering it. Hindsight relabeling, which supplies goal labels the demonstrations never contained, is the only imitation variant that clears the play policy at every threshold. PPO-TM is flat at 0% across all three thresholds in Figure 6(a): under the same 701K-frame budget as our pipeline, on-policy RL has not converged enough to reorient the cube at all, let alone reach a coarse threshold.

PPO-PM reaches 86.7% SR@0.4 rad after 344M counted frames. Counting both human and autonomous data, our 701K-frame pipeline reaches 72.7% at that threshold and overtakes PPO-PM at the coarser ones (91.4% SR@0.6 rad and 96.1% SR@0.8 rad versus 86.7%), with approximately 1/490 as many counted frames – Figure 6(b) plots the SR@0.6 rad case. This advantage is specific to coarse acquisition and to the stated frame accounting: at the tightest threshold in Figure 6(a), SR@0.4 rad, PPO-PM remains substantially stronger than Ours (86.7% versus 72.7%). PPO-unlimited is substantially stronger at every threshold and consumes at least 360M frames, roughly 1/510 of which our pipeline uses. Held instead to our own 701K-frame budget, PPO-TM has not learned to reorient the cube at all (0% at every threshold) – the cleanest statement of the sample-efficiency gap this comparison is meant to isolate.

3) Task C: Real-World Reorientation: Figure 7 summarizes the hardware results. the play policy obtains 20.9/38.3/54.1%, which is statistically indistinguishable from the same policy in simulation (15.6/36.7/54.7%, Figure 6): a goal-agnostic wanderer buys the same goal-neighborhood acquisition on both platforms. The remaining rows reproduce the supplied hardware summary, whose trial protocol and logs are not yet recovered; until they are, we call those values provisional SR estimates rather than calibrated success probabilities, and the gap between them and the play row is not protocol-matched. [TODO: Re-measure the random-walk and play-guided-MPC rows under the one-minute-window protocol so that every bar in this figure shares one definition.]

4) Task D: RGB Reorientation: Task D tests whether play-guided planning remains effective when the controller must infer object state from images. It uses the same initial-goal pairs, action space, episode horizon, proposal budget, and SR thresholds as Task C. The principal comparison is RGB world-model MPC against play-only RGB control and the privileged state-world-model controller. Tag-derived pose is used to score evaluation trajectories but must not enter the RGB controller. [TODO: Freeze the camera input and history, visual data budget, train/test scene split, lighting and occlusion variation, goal representation, prediction target, image augmentation, and whether tags provide training supervision.]

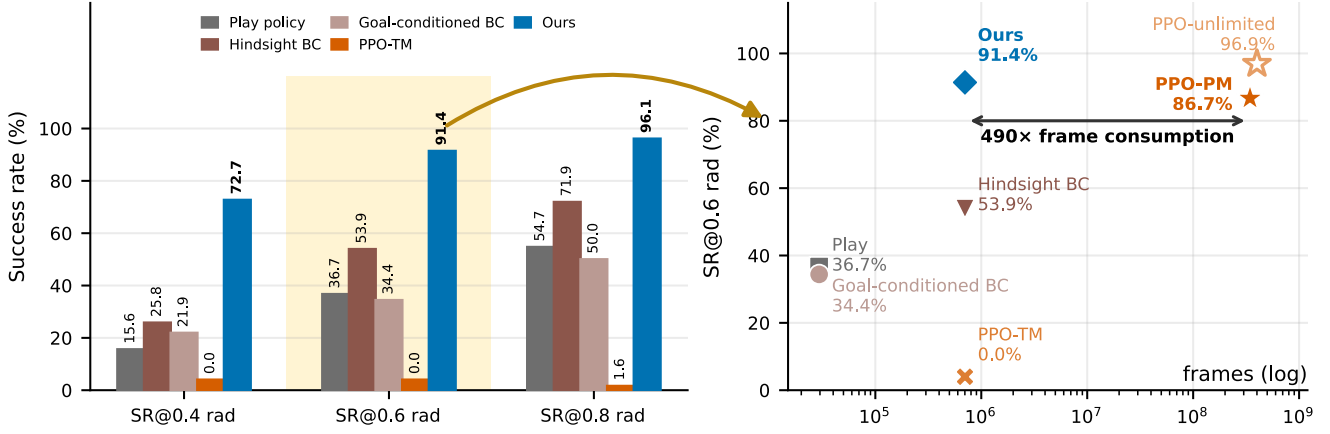


Fig. 6. **Simulated distant-goal reorientation** ($e_0 > 90^\circ$). (a) $SR@_\theta$, the fraction of $n = 128$ trials with $\min_t e_t \leq \theta$, under the same shared protocol. PPO-TM (timesteps matched) is PPO trained on the same 701K-frame budget as Ours/Hindsight BC and scores 0% at every threshold. Bold marks the best value per threshold. The highlighted $SR@0.6$ rad column is the slice plotted in (b), against counted training frames (log scale); Ours exceeds PPO-PM’s $SR@0.6$ rad with about $1/490$ as many counted frames. PPO-PM (performance matched) is a single 344M-frame checkpoint reaching comparable coarse performance to Ours; PPO-unlimited is the per-threshold median of five checkpoints from 360M frames onward and has no single frame count, so it is plotted at that floor (hollow marker).



Fig. 7. **Real-world reorientation success rates.** The sim-ours shows our real world performance matches our sim perf

TABLE II
RGB REORIENTATION RESULTS PLACEHOLDER. THE STATE-WM ROW IS A PRIVILEGED REFERENCE EVALUATED ON THE SAME TRIALS.

Controller	SR@0.4 rad	SR@0.6 rad	SR@0.8 rad
Play-only RGB	TBD	TBD	TBD
RGB WM + play-guided MPC	TBD	TBD	TBD
State WM + MPC (privileged)	TBD	TBD	TBD

D. Analysis

a) *Play-policy architectures.*: The relevant comparison is not open-loop imitation loss alone. A play policy must generate proposals quickly and cover useful state transitions. Table III compares the diffusion transformer used throughout the paper with a deterministic transformer of the same trunk (width 256, 6 blocks, 8 heads, 4-step patches, $H = 200$ with 50 executed steps) trained by mean-squared regression on the

TABLE III
PLAY-POLICY ARCHITECTURE ABLATION.

Policy	Avg. Rot. Speed (deg/s)	SO(3) Cov. (%)	Trans. Cov. (%)
Deterministic	57.8	87.6	55.4
Diffusion	124.9	98.8	84.2

same 29,474 human frames with the same 100K-step budget and validation-selected checkpoint, then rolled out for the same 512 seeded trials (identical initial orientations, 672K frames each). We score the collected play with three closed-loop statistics of the cube pose relative to the hand. Rotation speed is the cube’s in-hand path rotation per second of play, with reset teleports excluded. SO(3) coverage is the share of ten-degree Euler cells visited ($36 \times 18 \times 36 = 23,328$ cells; the pitch angle spans only 180°). Transition coverage treats fifteen-degree cells ($24 \times 12 \times 24 = 6,912$) as a state-transition graph with six directed unit moves per cell ($\pm 15^\circ$ in roll, pitch, or yaw; 40,320 transitions) and is the share of transitions realized by consecutive frames within a reset-free segment, so it rewards moving the cube in many distinct ways rather than merely reaching many poses. The two policies’ held-out action errors are similar (normalized command MSE 0.181 versus 0.192), but their closed-loop behavior diverges: the deterministic policy rotates the cube at less than half the rate, and although it still visits 88% of the orientation cells, it realizes only 55% of the possible unit transitions against 84% for the diffusion policy – it reaches most orientations, but through far fewer distinct motions, the signature of mode averaging. All values are recounted from the frozen corpora by `scripts/policy_arch_ablation.py`. [TODO: One seed per architecture; repeat with at least three seeds and add the flow-matching variant before drawing a general architecture conclusion.]

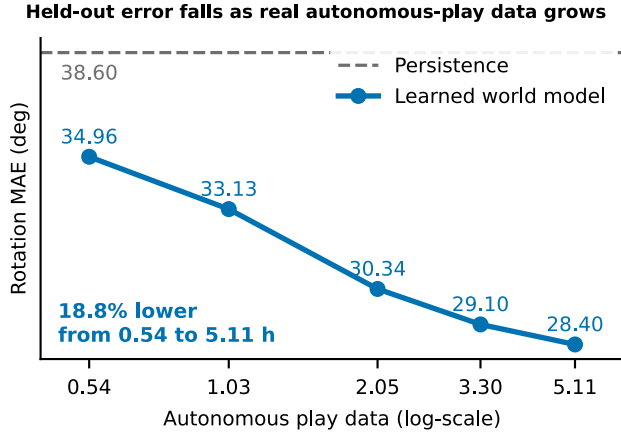


Fig. 8. **Held-out world-model error decreases with real autonomous play data.** Training subsets are nested and all evaluated settings are fixed. Each data budget has one training seed, so the figure establishes a controlled monotonic trend rather than a statistical scaling law. The persistence baseline is shown only as a reference.

b) Scaling autonomous play data.: We vary only the amount of real-robot autonomous play used to train the world model. The four nested subsets contain 0.54, 1.03, 2.05, and 2.34 hours (42,637, 82,219, 164,361, and 186,633 training windows). Architecture, normalization, optimization budget, and the 4,096-window held-out evaluation set are fixed. Increasing data from 0.54 to 2.34 hours monotonically reduces 50-step mean position MAE from 5.61 to 4.93 mm (12.1%) and rotation MAE from 34.96° to 30.06° (14.0%).

V. DISCUSSION RANDOMLY GENERATED BY CLAUDE, FEEL LIKE MOST THINGS CAN MERGE WITH RELATED WORKS

a) Ours versus play and behavior cloning.: Executing the play policy directly preserves its goal-agnostic nature but cannot deliberately target an arbitrary orientation. Hindsight BC introduces goal labels after collection, yet it can imitate only state-action-outcome tuples present in the recorded trajectories. MPC instead evaluates multiple counterfactual continuations from the current state. On the shared 701,474-frame benchmark, the evaluated hindsight-BC implementation obtains 25.8% SR@0.4 rad and 53.9% SR@0.6 rad, compared with 72.7% and 91.4% for our controller. This result does not isolate planning from model capacity, optimization, or compute, and does not generalize to every goal-conditioned imitation method. We include separately trained goal-conditioned BC as an additional same-task baseline.

b) Ours versus reinforcement learning.: DSP trades a small amount of human guidance and expensive online search for a lower counted-frame budget. At coarse goal acquisition, the 701k-frame system is comparable to PPO-Match, trained with 344M counted frames. Both PPO variants use a task-specific reward and a much larger logged dataset, while PPO-Best provides the stronger fine-accuracy, terminal-orientation, and drop-avoidance reference. The supported conclusion is resource-efficient goal acquisition under

the stated accounting and reusable dynamics, rather than overall superiority to RL.

c) Ours versus curiosity and random exploration.: Curiosity is a natural alternative for collecting goal-free experience, but no matched curiosity experiment is currently available. A fair comparison must fix interaction, hardware constraints, representation, and coverage metrics, then test both learned-model accuracy and downstream manipulation. The provisional hardware summary places joint and cube-space random walks below supervised play, but its protocol must be recovered before drawing a general exploration conclusion.

d) Limitations.: SR is an anytime acquisition metric; it does not require terminal accuracy or holding. Simulation respawns dropped cubes, which permits continued evaluation but masks failure severity. The real SR summary still lacks trial/session counts and a frozen success definition. The real data-scaling sweep has one training seed per budget. Push-T, RGB, curiosity, and the matched play-policy and world-model architecture ablations do not yet have frozen results in this draft. RGB reorientation is a required experiment, with the privileged-state controller serving as its reference.

e) Behavior foundation models as play policies.: The play policy is a modular proposal distribution. A behavior foundation model (BFM) could replace the task-specific behavior-cloned policy and provide broader object, contact, and semantic priors while leaving the world-model objective and MPC rule unchanged. This extension should be evaluated by closed-loop coverage, safety, proposal utility, and downstream planning after grounding the BFM in the robot's action space; open-loop likelihood alone is not sufficient.

VI. CONCLUSION RANDOMLY GENERATED BY CLAUDE, DO NOT READ

Dexterity from Supervised Play turns limited goal-free human play into autonomous contact-rich data, an action-conditioned world model, and goal-directed control through play-guided MPC. In simulation, the learned model supports arbitrary-goal reorientation with far fewer interactions than PPO-Match in counted training frames, and a controlled model-stack comparison shows improved goal acquisition under a fixed planner. On real-robot data, held-out prediction improves monotonically as autonomous play grows. A provisional hardware summary is consistent with a benefit from world-model search, but requires a frozen protocol and uncertainty before supporting a final task-success claim. The final evaluation includes both Push-T and RGB reorientation; their frozen quantitative results, together with curiosity and matched play-policy and world-model architecture comparisons, remain to be inserted.

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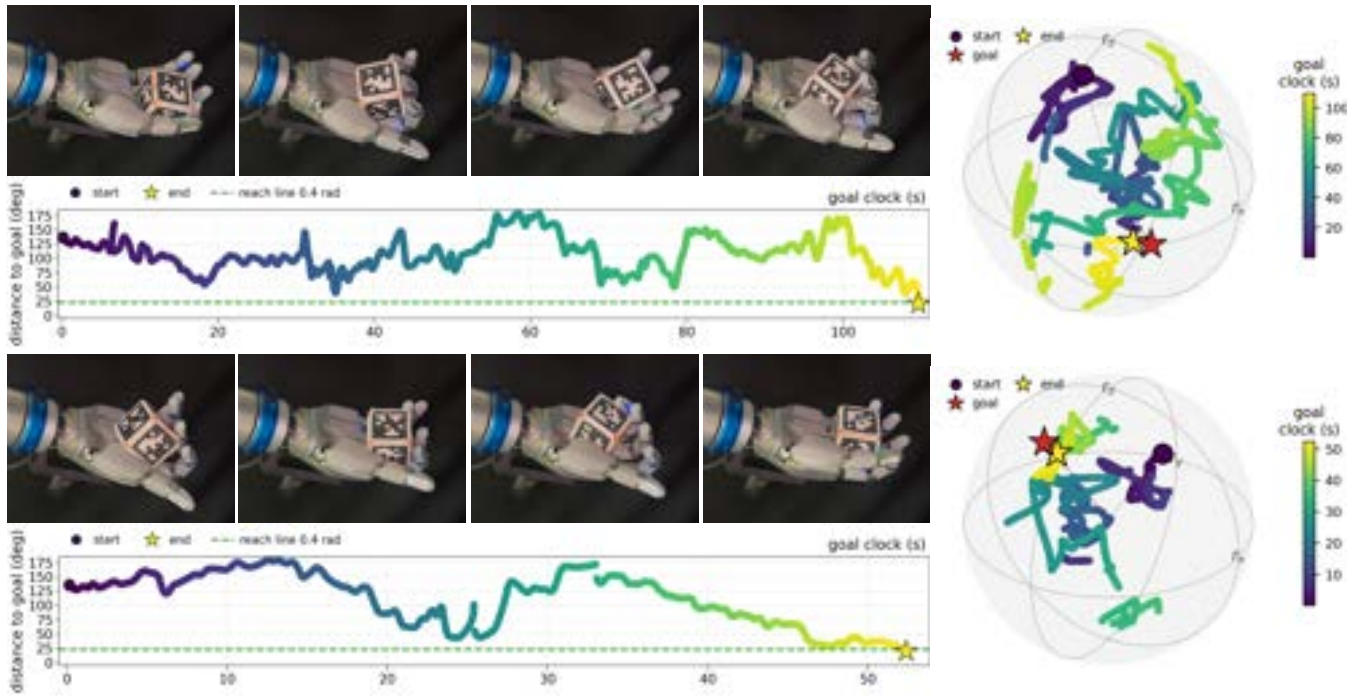


Fig. 9. Distance to goal. XXX

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